Casey Mathieson

Systems Engineer | Precision Automation & Machine Integration

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Professional Summary

Electro-mechanical systems engineer with 11+ years of experience designing, building, commissioning, and improving automation equipment, manufacturing processes, and precision-metrology systems. Brings end-to-end ownership from root-cause analysis and concept development through mechanical/electrical design, fabrication, test, deployment, and durable process documentation. Combines hands-on manufacturing knowledge with strong mechanical design, controls, calibration, and cross-functional delivery skills.

Technical Expertise

Systems engineering & integration: Requirements definition, system/subsystem integration, interfaces, risk/dependency mapping, root-cause analysis, and cross-functional technical coordination. **Precision metrology & validation:** Precision metrology, laser and sensor-head calibration, sensors/instrumentation, test-method development, repeatability, verification, and validation. **Mechanical design & manufacturing:** SolidWorks (CSWP), PDM, GD&T, complex assemblies, FEA/simulation, fixtures/nests, DFM/DFA, CNC/CAM/G-code, and fabrication. **Automation, controls & functional safety:** Motion and robotic systems, pneumatics, thermal control, AC/DC panel design, schematic review/redlines, motor control/VFDs, PLCs, and machine-safety design, programming, and validation. **Commissioning, quality & delivery:** System commissioning, FAT/SAT, PFMEA, SOPs/work instructions, process improvement, estimates/proposals, customer and field support, technician-ready documentation, and knowledge transfer. **AI-enabled engineering tooling & data workflows:** Agent-assisted tool development and validation; technical research, knowledge/file discovery, workflow automation, Python/MATLAB/TypeScript tools, and structured data.

Professional Experience

ZeroTouch Metrology / DWFritz Automation — Oregon

`Mar 2020–Present`

Systems Engineer, ZeroTouch · Sep 2024–Present

- Revived a stalled precision-stage qualification program, progressing from legacy-method audit and control-system migration through automated micrometer-scale analysis, validation testing, and controlled SOPs.
- Owned redesign, validation, and deployment of full-machine Golden Calibration across 9 systems, reducing the workflow from roughly 14 elapsed hours with two people to 2–3 hours with one person through automated data import, pass/fail decisions, and controlled exports; trained two field-service engineers.
- Across the ZeroTouch program, converted early-stage commissioning practices into a consistent workflow; built a 100+ page Confluence space with approximately 40 standard processes used throughout machine builds, supported 18 precision-metrology builds, and led 6 through end-to-end commissioning.
- Lead multidisciplinary troubleshooting across mechanical, electrical, controls, software, manufacturing, and field issues, converting findings into durable process, test-method, and documentation changes that reduce recurrence and streamline future diagnosis.

Systems Engineer, DWFritz · Jul 2024–Sep 2024

Mechanical Engineer, DWFritz · Mar 2020–Jul 2024

- Modernized legacy machine designs and developed mechanical solutions for conveyance, packaging, part handling, robotic end-of-arm tooling, precision fixtures, nests, and adjustment stages.
- Owned U.S. sensor-head calibration; personally calibrated more than 150 heads, validated calibration-system performance, redesigned the calibration system for improved repeatability and ease of use, qualified replacement gauge material and finish, created process documentation, and trained five colleagues.
- Improved metrology reliability through thermal R&D: identified sensor-head heat dissipation as a repeatability risk, validated cooling mitigations, and contributed to a condensation-prevention setpoint model using dew-point calculations, airflow assumptions, and thermoelectric cooling-unit fin-temperature test data.

Evergreen Advancements — Oregon

2023–Present

Founder / Product Systems Engineer · cascadia.studio · type-hype.com

- Independently architected and built a multi-brand product-operations platform using agentic workflows, integrating data models, asset preparation, pricing, marketplace publishing, analytics, and lifecycle controls.
- Built Cascadia Digital's Next.js/Firebase storefront with catalog search, checkout/download fulfillment, telemetry, responsive end-to-end testing, and fail-closed deployment.
- Built TypeHype, a zero-auth semantic font-discovery product with vision enrichment, pgvector retrieval and reranking, observability, cost controls, and canary validation across 250+ fonts and 500+ prewarmed discovery routes.

Sig Sauer Electro-Optics — Wilsonville, OR

Oct 2019–Feb 2020

Manufacturing Engineer

- Stabilized a production-bound six-robot, high-volume red-dot-sight assembly line by designing and retrofitting fault detection throughout the system, including part-presence, fluid-level, and cylinder-position sensing.
- Reduced line downtime by more than 65% and shifted operation from continuous support by two engineers to one technician with an engineer on call.

Production Robotics Inc. — San Leandro, CA

May 2015–Jul 2019

Senior Automation Engineer · Nov 2016–Jul 2019

Mechanical Design Engineer · May 2015–Nov 2016

- Led delivery of $30K–$1.5M purpose-built manufacturing and assembly systems from requirements through mechanical/electrical design, PFMEA, FAT/SAT, installation, commissioning, and production support.
- Led design, integration, validation, and commissioning of multi-robot vehicle assembly cells in under six weeks each, directing a four-person team of technicians and junior engineers and integrating vision, light curtains, LiDAR safety scanners, and facility control/data systems.
- Designed a FANUC end effector for thermal-bar placement and integrated vision-guided handling/fastening workflows for thermal-bar and Front Drive Unit mount installation, including conveyor-fed parts, pose correction, post-placement verification, bolt pickup, and automated fastening.
- Designed mechanical assemblies and ground-up AC/DC electrical panels, including 240/480 VAC power, motor controls/VFDs, and NFPA/NEC/NEMA-aligned schematics.

West Coast Surgical LLC — Half Moon Bay, CA

Oct 2014–May 2015

CNC Machinist

- Produced high-mix, low-volume medical-device work.
- Optimized milling setups, G-code, and workholding fixtures to meet tolerance requirements.

Shaffer's Offroad — Alameda, CA

Jun 2014–Oct 2014

CAD Designer / Fabricator

- Designed and manufactured custom vehicle components using SolidWorks, CNC plasma, press-brake, milling, and MIG-welding processes.

NASA Ames Research Center — Moffett Field, CA

Jun 2013–Oct 2014

Engineering Intern, Volunteer

- Contributed to R&D for a cable-actuated tensegrity rolling robot and led a senior-project team that prototyped a modular motorized cable-actuation system.

Education & Credentials

B.S., Mechanical Engineering, Mechatronics Focus — San Francisco State University, 2015

A.A., Economics; A.A., Marketing, Graphic Design Focus — Santa Barbara City College, 2011

Certified SOLIDWORKS Professional — Mechanical Design — Certificate C-QNHPJK9BLV, 2019